JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Education to Eternity

Basic Science & Humanities Department

2023-24 (Odd Semester)

CLASS TEST- I EXAMINATION

Course: B. Tech.

Subject: Advanced Physics **Subject Code:** CS1T002/AI1T002/DS1T002 **Max Marks: 20**

Year/Semester: 1st Semester (First Year) **Date: 25/09/2023** **Duration:- 1 Hr**

Instructions to the Students:

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

CO1: Describe the concept of LASER, optical fiber, types of semiconductors, PN junction diode characteristics, wave optics, electron Ballistics, quantum mechanics, various crystal structure parameters & X-rays.

CO2: Elaborate the types of LASER, optical fiber, Semiconductors, crystal structure, formation of Newton's ring, fringes in wedge shape thin film, effect of electric and magnetic field on motion of charge particle and significance of quantum mechanics.

CO3: Apply the concept of three and four level in LASER production, TIR in Optical fiber, classify the type of material based on current conduction, Bragg's law and X-ray diffraction, Interference for advanced application, illustrate the wave particle dualism of matter waves, motion and charged particle in E and B.

CO4: Analyze the behaviour of PN junction diode in FB and RB, compare the different types of LASER and optical fiber, correlate the motion of charged particles in uniform electric and magnetic fields for Bainbridge Mass spectrograph, the formation of fringes in thin film, behaviour of wave function and the types of crystal.

CO5: Justify physical significance of wave function, HUP, Schrodinger's wave equations, application of Hall effect, LASER & Optical Fibre, Wave Optics, Electron Ballistics and interpret he various crystal structure.

CO6: Design devices by using the concept of Laser, optical fibre, Electron ballistics, Semiconductor, crystals structure, wave optics and quantum mechanics.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	What is LASER stands for?	1/CO1	1
Q.2	What is level pumping scheme takes place in He-Ne LASER?	1/CO1	1
Q.3	What is Optical fibre?	1/CO1	1
Q.4	Explain the structure of Optical fibre.	2/CO2	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	Define the term i)Meta-stable state ii)Population Inversion	2/CO2	2
Q.6	Explain Stimulated Emission with suitable diagrams.	2/CO2	2
Q.7	Write any two applications of LASER.	2/CO2	2
Q.8	Explain Optical Resonator cavity.	2/CO2	2

SECTION C

1. **Solve any two questions.**

Q.9	Explain the Characteristics of LASER.	2/CO2	4
Q.10	Elaborate three level and four level pumping scheme.	3/CO3	4
Q.11	Analyze the working of Ruby LASER with energy level diagram.	4/CO4	4

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